Rudransh Singh

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EDUCATION

University of California, Berkeley May 2029
B.S Electrical Engineering & Computer Science

EXPERIENCE

Cofounder | Planetary Health July 2022 – Present

Co-founded a health-tech startup to enhance EHR (Electronic Health Record) security in under-resourced hospitals using blockchain. Collaborated on building the backend prototype and API for managing encrypted health records, and tested functionality through local development tools.

- Implemented a smart contract-based system for immutable health record storage
- Designed API layer in Express.js to facilitate client-side data access
- Published to IEEE CCCI 2024 - DOI: 10.1109/CCCI61916.2024.10736469
- Raised $110K in VC funding from Z-fellows and Angel investors.

Software Engineering | 10Spy June 2023 - March 2024

Collaborated with CEO to build a smart data-extraction tool using LangChain and OpenAI's GPT-4 API. Focused on API integration and prompt engineering.

- Integrated LangChain chains to pre-process and validate API inputs before generating responses
- Built a Python-based REST API to transform chatbot outputs into structured JSON using FastAPI, validated through schema testing in Pydantic
- Integrated LangChain chains to pre-process, rephrase, and contextually rewrite queries before being injected into OpenAI prompts
- Developed asynchronous request handling with asyncio to reduce latency across chained prompt operations

Machine Learning Research Summer 2022 - February 2025

Published four research papers at international IEEE conferences on implementations of machine learning. Trained CNN models to detect bias in generated images of professions and fine-tuned BERT models on tweets consisting of voter sentiment to predict the outcome of Prop-16 in California. Worked under mentors from UC Berkeley and UMD.

- Published in IEEE SaTML 2023 - DOI: 10.1109/ICAIC60265.2024.10433840
- Developed a stance detection model for tweets and applied the model to real elections
- Used BERT and ROBERTA models for prediction with 82.12% accuracy
- Research Paper published to IEEE and ICTC conferences 2022 - DOI: 10.1109/ICTC55196.2022.9952951